EXTENSION 2

Stage 6

Calculus (Ext2), C1 Integration (Ext2)

Recurrence Relations

Teacher: Nardene Dodd

Exam Equivalent Time: 136.5 minutes *(based on allocation of 1.5 minutes per mark)*

Questions

1. Calculus, EXT2 C1 2012 HSC 12c

For every integer $n \geq 0$ let $I_n = \int_1^{e^2} (\log_e x)^n \, dx$.

Show that for $n \geq 1$,

$$I_n = e^2 2^n - nI_{n-1}. \quad \textit{(3 marks)}$$

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2. Calculus, EXT2 C1 2021 HSC 13c

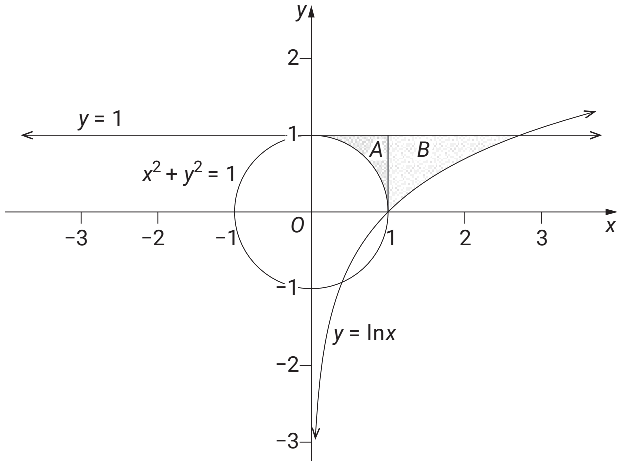
i. The integral I_n is defined by $I_n = \int_1^e (\ln x)^n \, dx$ for integers $n \geq 0$.

Show that $I_n = e - nI_{n-1}$ for $n \geq 1$. *(2 marks)*

ii. The diagram shows two regions.

Region A is bounded by $y = 1$ and $x^2 + y^2 = 1$ between $x = 0$ and $x = 1$.

Region B is bounded by $y = 1$ and $y = \ln x$ between $x = 1$ and $x = e$.



The volume of the solid created when the region between the curve $y = f(x)$ and the x-axis, between $x = a$ and $x = b$, is rotated about the x-axis is given by $V = \pi \int_a^b [f(x)]^2 \, dx$.

The volume of the solid of revolution formed when region A is rotated about the x-axis is V_A.

The volume of the solid of revolution formed when region B is rotated about the x-axis is V_B.

Using part (i), or otherwise, show that the ratio $V_A : V_B$ is $1:3$. *(4 marks)*

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3. Calculus, EXT2 C1 2018 HSC 14c

Let $I_n = \int_{-3}^0 x^n \sqrt{x+3} \, dx$ for $n = 0, 1, 2 \ldots$

i. Show that, for $n \geq 1$,

$$I_n = \frac{-6n}{3+2n} I_{n-1}. \quad \textit{(3 marks)}$$

ii. Find the value of I_2. *(2 marks)*

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4. Calculus, EXT2 C1 2019 HSC 15c

i. Show that $\displaystyle\int_0^1 \frac{x}{(x+1)^2}\,dx = \ln 2 - \frac{1}{2}$. *(2 marks)*

ii. Let $\displaystyle I_n = \int_0^1 \frac{x^n}{(x+1)^2}\,dx$.

Show that $\displaystyle I_n = \frac{1}{2(n-1)} - \frac{n}{n-1}I_{n-1}$ for $n \geq 2$. *(3 marks)*

iii. Evaluate I_3. *(2 marks)*

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5. Calculus, EXT2 C1 2022 HSC 14b

Let $\displaystyle J_n = \int_0^1 x^n e^{-x}\,dx$, where "n" is a non-negative integer.

i. Show that $J_0 = 1 - \dfrac{1}{e}$. *(1 mark)*

ii. Show that $J_n \leq \dfrac{1}{n+1}$. *(2 marks)*

iii. Show that $J_n = nJ_{n-1} - \dfrac{1}{e}$. *(2 marks)*

iv. Using parts (i) and (iii), show by mathematical induction, or otherwise, that for all $n \geq 1$,

$$J_n = n! - \frac{n!}{e}\sum_{r=0}^{n}\frac{1}{r!}$$ *(2 marks)*

v. Using parts (ii) and (iv) prove that $\displaystyle e = \lim_{n\to\infty}\sum_{r=0}^{n}\frac{1}{r!}$. *(1 mark)*

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6. Calculus, EXT2 C1 2014 HSC 12d

Let $\displaystyle I_n = \int_0^1 \frac{x^{2n}}{x^2+1}\,dx$, where n is an integer and $n \geq 0$.

i. Show that $I_0 = \dfrac{\pi}{4}$. *(1 mark)*

ii. Show that

$$I_n + I_{n-1} = \frac{1}{2n-1}.$$ *(2 marks)*

iii. Hence, or otherwise, find

$$\int_0^1 \frac{x^4}{x^2+1}\,dx.$$ *(2 marks)*

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7. Calculus, EXT2 C1 2002 HSC 2b

For $n = 0, 1, 2, \ldots$

let $\displaystyle I_n = \int_0^{\frac{\pi}{4}} \tan^n\theta\,d\theta$.

i. Show that $I_1 = \dfrac{1}{2}\ln 2$. *(1 mark)*

ii. Show that, for $n \geq 2$,

$$I_n + I_{n-2} = \frac{1}{n-1}.$$ *(3 marks)*

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8. Calculus, EXT2 C1 2008 HSC 3c

For $n \geq 0$, let $\displaystyle I_n = \int_0^{\frac{\pi}{4}} \tan^{2n}\theta\,d\theta$.

i. Show that for $n \geq 1$,

$$I_n = \frac{1}{2n-1} - I_{n-1}.$$ *(2 marks)*

ii. Hence, or otherwise, calculate I_3. *(2 marks)*

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9. Calculus, EXT2 C1 2023 HSC 15a

i. Let $\displaystyle J_n = \int_0^{\frac{\pi}{2}} \sin^n\theta\,d\theta$ where $n \geq 0$ is an integer.

Show that $J_n = \dfrac{n-1}{n}J_{n-2}$ for all integers $n \geq 2$. *(3 marks)*

ii. Let $\displaystyle I_n = \int_0^1 x^n(1-x)^n\,dx$ where n is a positive integer.

By using the substitution $x = \sin^2\theta$, or otherwise,

show that $\displaystyle I_n = \frac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} \sin^{2n+1}\theta\,d\theta$. *(4 marks)*

iii. Hence, or otherwise, show that $I_n = \dfrac{n}{4n+2}I_{n-1}$, for all integers $n \geq 1$. *(2 marks)*

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10. Calculus, EXT2 C1 2009 HSC 5b

For each integer $n \geq 0$, let

$$I_n = \int_0^1 x^{2n+1}e^{x^2}\,dx.$$

i. Show that for $n \geq 1$,

$$I_n = \frac{e}{2} - nI_{n-1}.$$ *(2 marks)*

ii. Hence, or otherwise, calculate I_2. *(2 marks)*

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11. Calculus, EXT2 C1 2013 HSC 13a

Let $I_n = \int_0^1 \left(1-x^2\right)^{\frac{n}{2}}\,dx$, where $n \geq 0$ is an integer.

i. Show that

$$I_n = \frac{n}{n+1} I_{n-2} \text{ for every integer } n \geq 2. \quad \textit{(3 marks)}$$

ii. Evaluate I_5. *(2 marks)*

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12. Calculus, EXT2 C1 2016 HSC 14b

Let $I_n = \int_0^1 \frac{x^n}{\left(x^2+1\right)^2}\,dx,$ for $n = 0, 1, 2, \ldots.$

i. Using a suitable substitution, show that $I_0 = \frac{\pi}{8} + \frac{1}{4}$. *(3 marks)*

ii. Show that $I_0 + I_2 = \frac{\pi}{4}$. *(1 mark)*

iii. Find I_4. *(3 marks)*

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13. Calculus, EXT2 C1 2017 HSC 15a

Let $I_n = \int_0^1 x^n \sqrt{1-x^2}\,dx$, for $n = 0, 1, 2, \ \ldots$

i. Find the value of I_1. *(1 mark)*

ii. Using integration by parts, or otherwise, show that for $n \geq 2$

$$I_n = \left(\frac{n-1}{n+2}\right) I_{n-2}. \quad \textit{(3 marks)}$$

iii. Find the value of I_5. *(1 mark)*

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14. Calculus, EXT2 C1 2020 HSC 16b

Let $I_n = \int_0^{\frac{\pi}{2}} \sin^{2n+1}(2\theta)\ d\theta,\ n = 0, 1, \ldots$

i. Prove that $I_n = \frac{2n}{2n+1} I_{n-1},\ n \geq 1$. *(3 marks)*

ii. Deduce that $I_n = \frac{2^{2n}(n!)^2}{(2n+1)!}$. *(3 marks)*

Let $J_n = \int_0^1 x^n (1-x)^n\,dx,\ n = 0, 1, 2, \ldots$

iii. Using the result of part (ii), or otherwise, show that $J_n = \frac{(n!)^2}{(2n+1)!}$. *(3 marks)*

iv. Prove that $\left(2^n n!\right)^2 \leq (2n+1)!$. *(2 marks)*

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15. Calculus, EXT2 C1 2006 HSC 7b

i. Let $I_n = \int_0^x \sec^n t\ dt$, where $0 \leq x \leq \frac{\pi}{2}$.

Show that $I_n = \frac{\sec^{n-2}x\tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$. *(3 marks)*

ii. Hence find the exact value of

$$\int_0^{\frac{\pi}{3}} \sec^4 t\ dt. \quad \textit{(2 marks)}$$

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16. Calculus, EXT2 C1 2011 HSC 8a

For every integer $m \geq 0$ let

$$I_m = \int_0^1 x^m \left(x^2-1\right)^5\,dx.$$

Prove that for $m \geq 2$

$$I_m = \frac{m-1}{m+11} I_{m-2}. \quad \textit{(3 marks)}$$

Worked Solutions

1. Calculus, EXT2 C1 2012 HSC 12c

$$I_n = \int_1^{e^2} (\log_e x)^n \, dx$$

$$u = (\log_e x)^n, \qquad v' = 1$$

$$u' = n(\log_e x)^{n-1} \times \frac{1}{x} \qquad v = x$$

$$I_n = \int_1^{e^2} 1 \cdot (\ln x)^n \, dx$$

$$= \left[x(\log_e x)^n\right]_1^{e^2} - \int_1^{e^2} \frac{n}{x}(\log_e x)^{n-1} \times x \, dx$$

$$= \left[e^2\left(\log_e e^2\right)^n - 1 \cdot \ln 1\right] - n\int_1^{e^2} (\log_e x)^{n-1} \, dx$$

$$= \left[e^2(2\log_e e)^n - 0\right] - nI_{n-1}$$

$$= e^2 2^n - nI_{n-1}$$

Worked Solutions

2. Calculus, EXT2 C1 2021 HSC 13c

i. $I_n = \displaystyle\int_1^e (\ln x)^n \, dx$ for $n \geq 0$

Show $I_n = e - nI_{n-1}$ for $n \geq 1$

Integrating by parts:

$$u = (\ln x)^n \qquad u' = \frac{n}{x}(\ln x)^{n-1}$$

$$v = x \qquad v' = 1$$

$$I_n = \left[x(\ln x)^n\right]_1^e - n\int (\ln x)^{n-1} \, dx$$

$$= \left[e(\ln e)^n - 1(\ln 1)^n\right] - nI_{n-1}$$

$$= e - nI_{n-1}$$

ii. $V_A =$ cylinder volume – hemisphere volume

$$= \pi\int_0^1 1 \, dx - \frac{1}{2} \times \frac{4}{3} \times \pi r^3$$

$$= \pi[x]_0^1 - \frac{2\pi}{3}$$

$$= \pi(1-0) - \frac{2\pi}{3}$$

$$= \frac{\pi}{3}$$

$$V_B = \pi\int_1^e 1 \, dx - \pi\int_1^e (\ln x)^2 \, dx$$

$$= \pi[x]_1^e - \pi I_2$$

$$= \pi(e-1) - \pi(e - 2I_1)$$

$$= \pi(e-1) - \pi[e - 2(e - I_0)]$$

$$= \pi(e-1) - \pi(2I_0 - e)$$

$$= \pi(e-1) - \pi[2(e-1) - e]$$

$$= \pi(e-1) - \pi(e-2)$$

$$= \pi e - \pi - \pi e + 2\pi$$

$$= \pi$$

$$\therefore V_A : V_B = \frac{\pi}{3} : \pi$$

$$= 1 : 3$$

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3. Calculus, EXT2 C1 2018 HSC 14c

i. $I_n = \int_{-3}^{0} x^n\sqrt{x+3}\ dx \quad \text{for} \quad n = 0, 1, 2\ldots$

$$u = x^n \qquad v' = (x+3)^{\frac{1}{2}}$$

$$u' = nx^{n-1} \qquad v = \frac{2}{3}(x+3)^{\frac{3}{2}}$$

$$I_n = \left[\frac{2}{3}x^n(x+3)^{\frac{3}{2}}\right]_{-3}^{0} - \frac{2}{3}\int_{-3}^{0} nx^{n-1}(x+3)\sqrt{x+3}\ dx$$

$$I_n = 0 - \frac{2n}{3}\int_{-3}^{0} x^n\sqrt{x+3} + 3x^{n-1}\sqrt{x+3}\ dx$$

$$I_n = -\frac{2n}{3}(I_n + 3I_{n-1})$$

$$I_n + \frac{2n}{3}I_n = -2nI_{n-1}$$

$$I_n\left(1 + \frac{2n}{3}\right) = -2nI_{n-1}$$

$$I_n\left(\frac{3+2n}{3}\right) = -2nI_{n-1}$$

$$\therefore I_n = \frac{-6n}{3+2n}I_{n-1} \quad \ldots \text{as required}$$

ii. $I_2 = -\frac{12}{7}I_1$

$$= -\frac{12}{7} \times -\frac{6}{5}I_0$$

$$= \frac{72}{35}\int_{-3}^{0}(x+3)^{\frac{1}{2}}dx$$

$$= \frac{72}{35} \times \frac{2}{3}\left[(x+3)^{\frac{3}{2}}\right]_{-3}^{0}$$

$$= \frac{48}{35}\left(3^{\frac{3}{2}}\right)$$

$$= \frac{144\sqrt{3}}{35}$$

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4. Calculus, EXT2 C1 2019 HSC 15c

i. Show $\int_0^1 \frac{x}{(x+1)^2}\ dx = \ln 2 - \frac{1}{2}$

Let $u = x+1 \Rightarrow x = u-1$

$\frac{du}{dx} = 1 \Rightarrow du = dx$

When $x = 1, \quad u = 2$

When $x = 0, \quad u = 1$

$$\int_0^1 \frac{x}{(x+1)^2}\ dx = \int_1^2 \frac{u-1}{u^2}\ du$$

$$= \int_1^2 \frac{1}{u}\ du - \int_1^2 \frac{1}{u^2}\ du$$

$$= [\ln u]_1^2 + \left[\frac{1}{u}\right]_1^2$$

$$= \ln 2 - \ln 1 + \frac{1}{2} - 1$$

$$= \ln 2 - \frac{1}{2}$$

ii. $I_u = \int_0^1 \frac{x^n}{(x+1)^2}\ dx$

$$u = x^n \qquad v' = \frac{1}{(x+1)^2}$$

$$u' = nx^{n-1} \qquad v = -\frac{1}{(x+1)}$$

$$I_n = [uv]_0^1 - \int_0^1 u'v\ dx$$

$$= \left[\frac{-x^n}{x+1}\right]_0^1 + \int_0^1 \frac{nx^{n-1}}{x+1}\ dx$$

$$= \left(-\frac{1}{2} - 0\right) + n\int_0^1 \frac{x^{n-1}(x+1)}{(x+1)^2}\ dx$$

$$= -\frac{1}{2} + n\int_0^1 \frac{x^n}{(x+1)^2} + \frac{x^{n-1}}{(x+1)^2}\ dx$$

$$= -\frac{1}{2} + nI_n + nI_{n-1}$$

$$nI_n - I_n = \frac{1}{2} - nI_{n-1}$$

$$I_n(n-1) = \frac{1}{2} - nI_{n-1}$$

$$\therefore I_n$$

$$= \frac{1}{2(n-1)} - \frac{n}{n-1} I_{n-1}$$

iii. $I_1 = \ln 2 - \frac{1}{2}$

$$I_2 = \frac{1}{2(2-1)} - \frac{2}{(2-1)} I_1$$

$$= \frac{1}{2} - 2I_1$$

$$= \frac{1}{2} - 2\ln 2 + 1$$

$$= \frac{3}{2} - 2\ln 2$$

$$\therefore I_3 = \frac{1}{4} - \frac{3}{2}\left(\frac{3}{2} - 2\ln 2\right)$$

$$= \frac{1}{4} - \frac{9}{4} + 3\ln 2$$

$$= 3\ln 2 - 2$$

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5. Calculus, EXT2 C1 2022 HSC 14b

i. $J_0 = \displaystyle\int_0^1 e^{-x}\, dx$

$$= \left[-e^{-x}\right]_0^1$$

$$= -e^{-1} + 1$$

$$= 1 - \frac{1}{e}$$

Mean mark (i) 93%.

ii. Show $J_n \le \dfrac{1}{n+1}$

♦♦ Mean mark (ii) 28%.

Note: $e^{-x} < 1$ for $x \in [0, 1]$

$$J_n = \int_0^1 x^n e^{-x}\, dx$$

$$\le \int_0^1 x^n\, dx$$

$$\le \frac{1}{n+1}\left[x^{n+1}\right]_0^1$$

$$\le \frac{1}{n+1}\left(1^{n+1} - 0\right)$$

$$\le \frac{1}{n+1} \quad \ldots \text{as required}$$

iii. Show $J_n = nJ_{n-1} - \dfrac{1}{e}$

$u = x^n \qquad v\prime = e^{-x}$

$u\prime = nx^{n-1} \qquad v = -e^{-x}$

$$J_n = \left[-x^n \cdot e^{-x}\right]_0^1 - \int_0^1 nx^{n-1} \cdot -e^{-x}\, dx$$

$$= \left(-1^n \cdot e^{-1} + 0^n e^0\right) + n\int_0^1 x^{n-1} \cdot e^{-x}\, dx$$

$$= nJ_{n-1} - \frac{1}{e}$$

iv. Prove $J_n = n! - \dfrac{n!}{e}\displaystyle\sum_{r=0}^{n} \frac{1}{r!}$ for $n \ge 0$

If $n = 0$:

LHS $= 1 - \dfrac{1}{e}$ (see part (i))

♦ Mean mark (iv) 50%.

RHS $= 0! - 0\dfrac{!}{e}\left(\dfrac{1}{0!}\right) = 1 - \dfrac{1}{e}(1) =$ LHS

$\therefore$ True for $n = 0$.

Assume true for $n = k$:

$$J_k = k! - \frac{k!}{e} \sum_{r=0}^{k} \frac{1}{r!}$$

Prove true for $n = k + 1$:

i.e. $J_{k+1} = (k+1)! - \dfrac{(k+1!)}{e} \displaystyle\sum_{r=0}^{k+1} \frac{1}{r!}$

$$\begin{aligned} J_{k+1} &= (k+1)J_k - \frac{1}{e} \quad \text{(using part (iii))} \\ &= (k+1)\left(k! - \frac{k!}{e}\sum_{r=0}^{k}\frac{1}{r!}\right) - \frac{1}{e} \\ &= (k+1)! - \frac{(k+1)!}{e}\sum_{r=0}^{k}\frac{1}{r!} - \frac{1}{e} \times \frac{(k+1)!}{(k+1)!} \\ &= (k+1)! - \frac{(k+1)!}{e}\left(\sum_{r=0}^{k}\frac{1}{r!} + \frac{1}{(k+1)!}\right) \\ &= (k+1)! - \frac{(k+1)!}{e}\left(\sum_{r=0}^{k+1}\frac{1}{r!}\right) \end{aligned}$$

$\Rightarrow$ True for $n = k + 1$

$\therefore$ Since true for $n = 1$, by PMI, true for integers $n \geq 1$

v. $0 \leq J_n \leq \dfrac{1}{n+1}$ (part (ii))

♦♦ Mean mark (v) 34%.

$$\lim_{n\to\infty} \frac{1}{n+1} = 0 \Rightarrow \lim_{n\to\infty} J_n = 0$$

Using part (iv):

$$\begin{aligned} \frac{J_n}{n!} &= 1 - \frac{1}{e}\sum_{r=0}^{n}\frac{1}{r!} \\ \frac{1}{e}\sum_{r=0}^{n}\frac{1}{r!} &= 1 - \frac{J_n}{n!} \\ \sum_{r=0}^{n}\frac{1}{r!} &= e - \frac{eJ_n}{n!} \\ \lim_{n\to\infty}\left(\sum_{r=0}^{n}\frac{1}{r!}\right) &= \lim_{n\to\infty}\left(e - \frac{eJ_n}{n!}\right) \\ &= e - 0 \\ &= e \end{aligned}$$

6. Calculus, EXT2 C1 2014 HSC 12d

i. $I_n = \displaystyle\int_0^1 \frac{x^{2n}}{x^2+1}\,dx, \quad n \geq 0$

$$\begin{aligned} I_0 &= \int_0^1 \frac{x^0}{x^2+1}\,dx \\ &= \left[\tan^{-1}x\right]_0^1 \\ &= \tan^{-1}1 - \tan^{-1}0 \\ &= \frac{\pi}{4} \end{aligned}$$

ii. $I_{n-1} = \displaystyle\int_0^1 \frac{x^{2n-2}}{x^2+1}\,dx$

♦ *Mean mark 41%.*
STRATEGY: The denominator of the proof, $(2n-1)$, should flag the potential existence of x^{2n-2} in the integral.

$$I_n = \int_0^1 \frac{x^{2n}}{x^2+1}\,dx$$

$$\begin{aligned} I_n + I_{n-1} &= \int_0^1 \frac{x^{2n}}{x^2+1}\,dx + \int_0^1 \frac{x^{2n-2}}{x^2+1}\,dx \\ &= \int_0^1 \frac{x^{2n}+x^{2n-2}}{x^2+1}\,dx \\ &= \int_0^1 \frac{x^{2n-2}(x^2+1)}{x^2+1}\,dx \\ &= \int_0^1 x^{2n-2}\,dx \\ &= \left[\frac{x^{2n-1}}{2n-1}\right]_0^1 \\ &= \frac{1}{2n-1} \end{aligned}$$

iii. $I_2 = \displaystyle\int_0^1 \frac{x^4}{x^2+1}\,dx$

$$I_1 = \int_0^1 \frac{x^2}{x^2+1}\,dx$$

$$\begin{aligned} I_n + I_{n-1} &= \frac{1}{2n-1} \\ I_2 + I_1 &= \frac{1}{2(2)-1} = \frac{1}{3} \quad \ldots (1) \\ I_1 + I_0 &= \frac{1}{2(1)-1} = 1 \\ I_1 + \frac{\pi}{4} &= 1 \\ \therefore I_1 & \end{aligned}$$

$$= 1 - \frac{\pi}{4}$$

Substitute into (1)

$$I_2 + 1 - \frac{\pi}{4} = \frac{1}{3}$$

$$\therefore I_2 = \frac{\pi}{4} - \frac{2}{3}$$

7. Calculus, EXT2 C1 2002 HSC 2b

i. $$I_n = \int_0^{\frac{\pi}{4}} \tan^n\theta \, d\theta$$

$$I_1 = \int_0^{\frac{\pi}{4}} \tan\theta \, d\theta$$

$$= [-\ln\cos x]_0^{\frac{\pi}{4}}$$

$$= \left[-\ln\cos\frac{\pi}{4} + \ln\cos 0\right]$$

$$= -\ln 2^{-\frac{1}{2}}$$

$$= \frac{1}{2}\ln 2$$

ii. $$I_n = \int_0^{\frac{\pi}{4}} \tan^{n-2}\theta \cdot \tan^2\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2}\theta \cdot \left(\sec^2\theta - 1\right) d\theta$$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2}\theta \cdot \sec^2\theta \, d\theta - \int_0^{\frac{\pi}{4}} \tan^{n-2}\theta \, d\theta$$

Let $u = \tan\theta$

$$\frac{du}{d\theta} = \sec^2\theta \Rightarrow du = \sec^2\theta \, d\theta$$

When $\theta = \frac{\pi}{4}, \quad u = 1$

$\theta = 0, \quad u = 0$

$$I_n = \int_0^1 u^{n-2} \, du - I_{n-2}$$

$$= \left[\frac{1}{n-1} \cdot u^{n-1}\right]_0^1 - I_{n-2}$$

$$= \frac{1}{n-1} - I_{n-2}$$

$$\therefore I_n + I_{n-2} = \frac{1}{n-1} \quad (n \geq 2)$$

8. Calculus, EXT2 C1 2008 HSC 3c

i. $$I_n = \int_0^{\frac{\pi}{4}} \tan^{2n-2}\theta \cdot \tan^2\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{4}} \tan^{2n-2}\theta \cdot \left(\sec^2\theta - 1\right) d\theta$$

$$= \int_0^{\frac{\pi}{4}} \tan^{2n-2}\theta \cdot \sec^2\theta \, d\theta - \int_0^{\frac{\pi}{4}} \tan^{2n-2}\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{4}} \tan^{2n-2}\theta \cdot \sec^2\theta \, d\theta - I_{n-1}$$

Let $u = \tan\theta$

$$\frac{du}{d\theta} = \sec^2\theta \Rightarrow du = \sec^2\theta \, d\theta$$

When $\theta = \frac{\pi}{4}, \quad u = 1$

$\theta = 0, \quad u = 0$

$$I_n = \int_0^1 u^{2n-2} \, du - I_{n-1}$$

$$= \left[\frac{1}{2n-1} \cdot u^{2n-1}\right]_0^1 - I_{n-1}$$

$$= \frac{1}{2n-1}\left(1^{2n-1} - 0\right) - I_{n-1}$$

$$= \frac{1}{2n-1} - I_{n-1}$$

ii. $$I_0 = \int_0^{\frac{\pi}{4}} d\theta = \frac{\pi}{4}$$

$$I_3 = \frac{1}{5} - I_2$$

$$= \frac{1}{5} - \left(\frac{1}{3} - I_1\right)$$

$$= \frac{1}{5} - \frac{1}{3} + (1 - I_0)$$

$$= \frac{13}{15} - \frac{\pi}{4}$$

$$= \frac{52 - 15\pi}{60}$$

9. Calculus, EXT2 C1 2023 HSC 15a

i. $J_n = \int_0^{\frac{\pi}{2}} \sin^n \theta \, d\theta$

$= \int_0^{\frac{\pi}{2}} \sin^{n-1} \theta \cdot \sin \theta \, d\theta$

Integrating by parts:

$u = \sin^{n-1}\theta \qquad u' = (n-1)\cos\theta \cdot \sin^{n-2}\theta$

$v = -\cos\theta \qquad v' = \sin\theta$

$J_n = \left[-\cos\theta \cdot \sin^{n-1}\theta\right]_0^{\frac{\pi}{2}} + (n-1)\int_0^{\frac{\pi}{2}} \cos^2\theta \cdot \sin^{n-2}\theta \, d\theta$

$= (n-1)\int_0^{\frac{\pi}{2}} (1-\sin^2\theta)\sin^{n-2}\theta \, d\theta$

$= (n-1)\int_0^{\frac{\pi}{2}} \sin^{n-2}\theta - \sin^n\theta \, d\theta$

$= (n-1)J_{n-2} - (n-1)J_n$

$J_n + (n-1)J_n = (n-1)J_{n-2}$

$J_n(1+n-1) = (n-1)J_{n-2}$

$J_n = \dfrac{n-1}{n} J_{n-2}$

ii. $I_n = \int_0^1 x^n(1-x)^n \, dx$

♦♦ Mean mark (ii) 34%.

Let $x = \sin^2\theta$

$\dfrac{dx}{d\theta} = 2\sin\theta\cos\theta \Rightarrow dx = 2\sin\theta\cos\theta \, d\theta$

When $x = 1,\ \theta = \dfrac{\pi}{2};\ \ x = 0,\ \theta = 0$

$I_n = \int_0^{\frac{\pi}{2}} \sin^{2n}\theta(1-\sin^2\theta)^n \cdot 2\sin\theta\cos\theta \, d\theta$

$= 2\int_0^{\frac{\pi}{2}} \sin^{2n+1}\theta \cdot \cos^{2n+1}\theta \, d\theta$

$= \dfrac{2}{2^{2n+1}}\int_0^{\frac{\pi}{2}} (2\sin\theta\cos\theta)^{2n+1} d\theta$

$= \dfrac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} \sin^{2n+1}(2\theta) \, d\theta$

Let $u = 2\theta$

$\dfrac{du}{d\theta} = 2 \Rightarrow \dfrac{du}{2} = d\theta$

When $\theta = \dfrac{\pi}{2},\ u = \pi;\ \theta = 0,\ u = 0$

$I_n = \dfrac{1}{2^{2n+1}}\int_0^{\pi} \sin^{2n+1} u \, du$

Since $\sin^{2n+1} u$ is symmetrical about $u = \dfrac{\pi}{2}$

$\Rightarrow \int_0^{\pi} \sin^{2n+1} u \, du = 2\int_0^{\frac{\pi}{2}} \sin^{2n+1} u \, du$

$I_n = \dfrac{1}{2^{2n+1}} \cdot 2\int_0^{\frac{\pi}{2}} \sin^{2n+1} u \, du$

$= \dfrac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} \sin^{2n+1} u \, du$

Which can be expressed as:

$I_n = \dfrac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} \sin^{2n+1}\theta \, d\theta$

iii. $I_n = \dfrac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} \sin^{2n+1}\theta \, d\theta$

♦♦ Mean mark (iii) 28%.

$= \dfrac{1}{2^{2n}} J_{2n+1}$ (by definition)

$= \dfrac{1}{2^{2n}} \cdot \underbrace{\dfrac{2n}{2n+1} \cdot J_{2n+1}}_{\text{using part i}}$

$= \dfrac{n}{2(2n+1)} \cdot \dfrac{2^2}{2^{2n}} \cdot J_{2n-1}$

$= \dfrac{n}{4n+2} \cdot \underbrace{\dfrac{1}{2^{2n-2}} \cdot J_{2n-1}}_{=I_{n-1}\ \text{(using part ii)}}$

$= \dfrac{n}{4n+2} I_{n-1}$

10. Calculus, EXT2 C1 2009 HSC 5b

i. Integrating by parts:

$$u = x^{2n}, \qquad u\prime = 2nx^{2n-1}$$

$$v\prime = xe^{x^2}, \qquad v = \frac{1}{2}e^{x^2}$$

$$\begin{aligned}\therefore I_n &= \int_0^1 x^{2n+1}e^{x^2}\ dx \\ &= \int_0^1 x^{2n}xe^{x^2}\ dx \\ &= \left[\frac{x^{2n}e^{x^2}}{2}\right]_0^1 - \int_0^1 \frac{2nx^{2n}e^{x^2}}{2}\ dx \\ &= \frac{e}{2} - n\int_0^1 x^{2n}e^{x^2}\ dx \\ &= \frac{e}{2} - nI_{n-1}\end{aligned}$$

ii. $I_2 = \dfrac{e}{2} - 2I_1$

$$I_1 = \frac{e}{2} - I_0$$

$$\begin{aligned}I_0 &= \int_0^1 xe^{x^2}\ dx \\ &= \left[\frac{e^{x^2}}{2}\right]_0^1 \\ &= \frac{e-1}{2}\end{aligned}$$

$$I_1 = \frac{e}{2} - \frac{e-1}{2} = \frac{1}{2}$$

$$\begin{aligned}\therefore I_2 &= \frac{e}{2} - 2 \times \frac{1}{2} \\ &= \frac{e-2}{2}\end{aligned}$$

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11. Calculus, EXT2 C1 2013 HSC 13a

i. $I_n = \displaystyle\int_0^1 \left(1-x^2\right)^{\frac{n}{2}}\ dx$

$$\begin{aligned}\text{Let}\quad u &= \left(1-x^2\right)^{\frac{n}{2}} \\ u\prime &= \frac{n}{2}\left(1-x^2\right)^{\frac{n}{2}-1}(-2x) \\ &= -nx\left(1-x^2\right)^{\frac{n}{2}-1} \\ v &= x \\ v\prime &= 1\end{aligned}$$

♦ *Mean mark 46%.*
STRATEGY TIP: Attempting to prove this relationship using induction proved unsuccessful and time consuming for some students.

$$\begin{aligned}I_n &= \left[x\left(1-x^2\right)^{\frac{n}{2}}\right]_0^1 - \int_0^1 -nx\left(1-x^2\right)^{\frac{n}{2}-1}x\ dx \\ &= 0 + n\int_0^1 x^2\left(1-x^2\right)^{\frac{n}{2}-1}\ dx \\ &= n\int_0^1 \left(x^2-1+1\right)\left(1-x^2\right)^{\frac{n}{2}-1}\ dx \\ &= n\int_0^1 1\left(1-x^2\right)^{\frac{n}{2}-1}\ dx - n\int_0^1 \left(1-x^2\right)\left(1-x^2\right)^{\frac{n}{2}-1}\ dx \\ &= n\int_0^1 \left(1-x^2\right)^{\frac{n-2}{2}}\ dx - n\int_0^1 \left(1-x^2\right)^{\frac{n}{2}}\ dx \\ &= nI_{n-2} - nI_n\end{aligned}$$

$$\therefore (n+1)I_n = nI_{n-2}$$

$$I_n = \frac{n}{n+1}I_{n-2} \quad \text{where} \quad n \geq 2$$

STRATEGY TIP: Realising that the integral I_1 is equal to the area of a quarter unit circle saved valuable time.

ii. $I_5 = \dfrac{5}{6}I_3 = \dfrac{5}{6} \times \dfrac{3}{4}\ I_1$

$$\begin{aligned}I_1 &= \int_0^1 \left(1-x^2\right)^{\frac{1}{2}}\ dx \\ &= \frac{1}{4} \times \pi \times 1^2 \quad \text{(1st quadrant of a circle, radius 1)} \\ &= \frac{\pi}{4}\end{aligned}$$

$$\begin{aligned}I_5 &= \frac{5}{6} \times \frac{3}{4} \times \frac{\pi}{4} \\ &= \frac{5\pi}{32}\end{aligned}$$

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12. Calculus, EXT2 C1 2016 HSC 14b

i. $I_n = \int_0^1 \frac{x^n}{(x^2+1)^2} dx, \quad n \geq 0$

Let $x = \tan\theta$

$dx = \sec^2\theta \; d\theta$

When $x = 0, \;\; \theta = 0$

$x = 1, \;\; \theta = \frac{\pi}{4}$

$$\therefore I_0 = \int_0^{\frac{\pi}{4}} \frac{\sec^2\theta}{\left(\tan^2\theta + 1\right)^2} \; d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sec^2\theta}{\sec^4\theta} \; d\theta$$

$$= \int_0^{\frac{\pi}{4}} \cos^2\theta \; d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{2}(1 + \cos 2\theta) \; d\theta$$

$$= \frac{1}{2}\left[x + \frac{1}{2}\sin 2\theta\right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2}\left[\left(\frac{\pi}{4} + \frac{1}{2} \cdot \sin\frac{\pi}{2}\right) - 0\right]$$

$$= \frac{\pi}{8} + \frac{1}{4}$$

ii. $I_0 + I_2 = \int_0^1 \frac{1}{(x^2+1)^2} \; dx + \int_0^1 \frac{x^2}{(x^2+1)^2} \; dx$

$$= \int_0^1 \frac{x^2+1}{(x^2+1)^2} \; dx$$

$$= \int_0^1 \frac{1}{x^2+1} \; dx$$

$$= \left[\tan^{-1}x\right]_0^1$$

$$= \frac{\pi}{4}$$

iii. $I_4 = \int_0^1 \frac{x^4}{x^2+1} \; dx$

♦ *Mean mark part (iii) 49%.*

$$= \int_0^1 \frac{x^4-1}{(x^2+1)^2} \; dx + \int_0^1 \frac{1}{(x^2+1)^2}$$

$$= \int_0^1 \frac{(x^2+1)(x^2-1)}{(x^2+1)^2} \; dx + I_0$$

$$= \int_0^1 \frac{x^2-1}{x^2+1} \; dx + I_0$$

$$= \int_0^1 \left(1 - \frac{2}{x^2+1}\right) \; dx + I_0$$

$$= \left[x - 2\tan^{-1}x\right]_0^1 + \frac{\pi}{8} + \frac{1}{4}$$

$$= \left[\left(1 - 2 \cdot \frac{\pi}{4}\right) - 0\right] + \frac{\pi}{8} + \frac{1}{4}$$

$$= \frac{5}{4} - \frac{3\pi}{8}$$

13. Calculus, EXT2 C1 2017 HSC 15a

i. $I_1 = \int_0^1 x^1\sqrt{1-x^2}\ dx$

$$= -\frac{1}{2}\int_0^1 -2x\sqrt{1-x^2}\ dx$$

$$= -\frac{1}{2}\left[\frac{2}{3}\left(1-x^2\right)^{\frac{3}{2}}\right]_0^1$$

$$= -\frac{1}{2}\left[0-\frac{2}{3}(1)\right]$$

$$= \frac{1}{3}$$

ii. $I_n = \int_0^1 x^n\sqrt{1-x^2}\ dx$

♦ *Mean mark 48%.*

Integrating by parts:

$u = x^{n-1}$ $\qquad dv = x\sqrt{1-x^2}\ dx$

$du = (n-1)x^{n-2}\ dx$ $\qquad v = -\frac{1}{3}\left(1-x^2\right)^{\frac{3}{2}}$

$$I_n = \left[x^{n-1}\times -\frac{1}{3}\left(1-x^2\right)^{\frac{3}{2}}\right]_0^1 + \frac{1}{3}\int_0^1 \left(1-x^2\right)^{\frac{3}{2}}(n-1)x^{n-2}\ dx$$

$$= 0 + \frac{n-1}{3}\int_0^1 x^{n-2}\left(1-x^2\right)\cdot\sqrt{1-x^2}\ dx$$

$$= \frac{n-1}{3}\int_0^1 x^{n-2}\sqrt{1-x^2}\ dx - \frac{n-1}{3}\int_0^1 x^n\sqrt{1-x^2}\ dx$$

$$= \frac{n-1}{3}\cdot I_{n-2} - \frac{n-1}{3}\cdot I_n$$

$$\therefore I_n + \frac{n-1}{3}\cdot I_n = \frac{n-1}{3}\cdot I_{n-2}$$

$$I_n\left(1+\frac{n-1}{3}\right) = \frac{n-1}{3}\cdot I_{n-2}$$

$$I_n\left(\frac{n+2}{3}\right) = \frac{n-1}{3}\cdot I_{n-2}$$

$$\therefore I_n = \left(\frac{n-1}{n+2}\right)\cdot I_{n-2}\ \ldots \text{as required.}$$

iii. $I_5 = \frac{4}{7}\times I_3$

$$= \frac{4}{7}\times\frac{2}{5}\times I_1$$

$$= \frac{4}{7}\times\frac{2}{5}\times\frac{1}{3}\ \text{(part (i))}$$

$$= \frac{8}{105}$$

14. Calculus, EXT2 C1 2020 HSC 16b

♦ Mean mark (i) 48%.

i. Prove $I_n = \dfrac{2n}{2n+1} I_{n-1},\ n \geq 1$

$$I_n = \int_0^{\frac{\pi}{2}} \sin^{2n}(2\theta) \cdot \sin(2\theta)\ d\theta$$

Integrating by parts:

$u = \sin^{2n}(2\theta)$ $\quad u' = 2n\sin^{2n-1}(2\theta)\ \times -\dfrac{1}{2}\cos(2\theta)$

$v = -\dfrac{1}{2}\cos(2\theta)$ $\quad v' = \sin 2\theta$

$$I_n = \left[\sin^{2n}(2\theta) \cdot -\frac{1}{2}\cos(2\theta)\right]_0^{\frac{\pi}{2}} - 2n\int_0^{\frac{\pi}{2}} \sin^{2n-1}(2\theta) \cdot 2\cos(2\theta) \cdot -\frac{1}{2}\cos(2\theta)\ d\theta$$

$$I_n = 0 + 2n\int_0^{\frac{\pi}{2}} \sin^{2n-1}(2\theta) \cdot \cos^2(2\theta)\ d\theta$$

$$I_n = 2n\int_0^{\frac{\pi}{2}} \sin^{2n-1}(2\theta)\left(1 - \sin^2(2\theta)\right)\ d\theta$$

$$I_n = 2n\int_0^{\frac{\pi}{2}} \sin^{2n-1}(2\theta) - \sin^{2n+1}(2\theta)\ d\theta$$

$$I_n = 2n(I_{n-1} - I_n)$$

$$I_n + 2n\ I_n = 2nI_{n-1}$$

$$I_n(2n+1) = 2nI_{n-1}$$

$$\therefore I_n = \frac{2n}{2n+1} I_{n-1}$$

♦ Mean mark (ii) 36%.

ii. $I_0 = \displaystyle\int_0^{\frac{\pi}{2}} \sin(2\theta)\ d\theta$

$$= \left[-\frac{1}{2}\cos(2\theta)\right]_0^{\frac{\pi}{2}}$$

$$= \left(-\frac{1}{2}\cos\pi + \frac{1}{2}\cos 0\right)$$

$$= 1$$

$$I_n = \frac{2n}{2n+1} I_{n-1}$$

$$I_{n-1} = \frac{2(n-1)}{2n-1} I_{n-2}$$

$$\vdots$$

$$I_1 = \frac{2}{3} I_0$$

$$I_n = \frac{2n}{2n+1} \times \frac{2(n-1)}{2n-1} \times \frac{2(n-2)}{2n-3} \times \ldots \times \frac{2}{3} \times 1$$

$$= \frac{2n}{2n+1} \times \frac{2n}{2n} \times \frac{2(n-1)}{2n-1} \times \frac{2(n-1)}{2n-2} \times \ldots \times \frac{2}{3} \times \frac{2}{2} \times 1$$

$$= \frac{2^n(n \times (n-1) \times .. \times 1) \times 2^n(n \times (n-1) \times \ldots \times 1)}{(2n+1)!}$$

$$= \frac{2^{2n}(n!)^2}{(2n+1)!}$$

♦♦♦ Mean mark (iii) 16%.

iii. $J_n = \displaystyle\int_0^1 x^n(1-x)^n\ dx, \quad n = 0, 1, 2, \ldots$

Let $x = \sin^2\theta$

$$\frac{dx}{d\theta} = 2\sin\theta\ \cos\theta\ \Rightarrow\ dx = 2\sin\theta\ \cos\theta\ d\theta$$

When $x = 0\ ,\quad \theta = 0$

$x = 1\ ,\quad \theta = \dfrac{\pi}{2}$

$$J_n = \int_0^{\frac{\pi}{2}} \left(\sin^2\theta\right)^n \left(1 - \sin^2\theta\right)^n \cdot 2\sin\theta\ \cos\theta\ d\theta$$

$$= \int_0^{\frac{\pi}{2}} \sin^{2n}\theta\ \cos^{2n}\theta \cdot \sin(2\theta)\ d\theta$$

$$= \frac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} 2^{2n}\sin^{2n}\theta\ \cos^{2n}\theta \cdot \sin(2\theta)\ d\theta$$

$$= \frac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} \sin^{2n}(2\theta) \cdot \sin(2\theta)\ d\theta$$

$$= \frac{1}{2^{2n}}\int_0^{\frac{\pi}{2}} \sin^{2n+1}(2\theta)\ d\theta$$

$$= \frac{1}{2^{2n}} \cdot \frac{2^{2n}(n!)^2}{(2n+1)!} \quad \text{(using part (ii))}$$

$$= \frac{(n!)^2}{(2n+1)!}$$

♦♦♦ Mean mark (iv) 10%.

iv. If $I_n \leq 1$,

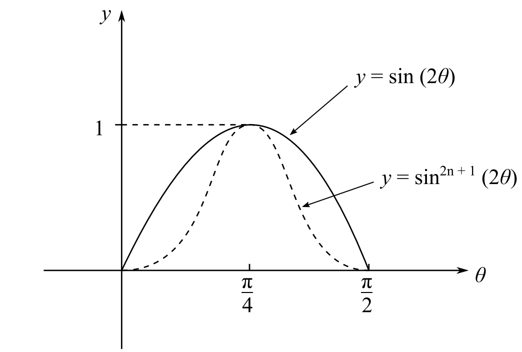
$$2^{2n}(n!)^2 \leq (2n+1)!$$

$$(2^n n!)^2 \leq (2n+1)!$$

Show $I_n \leq 1$:

Consider the graphs

$y=\sin(2\theta)$ and $y=\sin^{2n+1}(2\theta)$ for $0\le\theta\le\dfrac{\pi}{2}$



$$\int_0^{\frac{\pi}{2}}\sin(2\theta)=\left[-\frac{1}{2}\cos(2\theta)\right]_0^{\frac{\pi}{2}}$$
$$=-\frac{1}{2}\cos\,\pi+\frac{1}{2}\cos\,0$$
$$=1$$

$y=\sin(2\theta)\ \Rightarrow$ Range $[0,1]$ for $\theta\in\left[0,\dfrac{\pi}{2}\right]$

$\sin^{2n+1}(2\theta)\le\sin(2\theta)$ for $\theta\in\left[0,\dfrac{\pi}{2}\right]$

$$\sin^{2n+1}(2\theta)\le 1$$
$$I_n\le 1$$
$$\therefore (2^n n!)^2\le(2n+1)!$$

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15. Calculus, EXT2 C1 2006 HSC 7b

STRATEGY: The choice of $u=\sec^{n-2}t$ and $v=\sec^2 t$ was *extremely important* in being able to answer this question. Take note!

i. $I_n=\displaystyle\int_0^x \sec^n t\ dt,\quad 0\le x<\frac{\pi}{2}$

$$=\int_0^x \sec^{n-2}t\sec^2 t\ dt$$

Integrating by parts

$$u=\sec^{n-2}t,\qquad u\prime=(n-2)\sec^{n-2}t\tan\,t$$
$$v=\tan\,t,\qquad v\prime=\sec^2 t$$

$$I_n=\left[\tan t\sec^{n-2}t\right]_0^x-\int_0^x \tan t(n-2)\sec^{n-3}t\sec t\tan t\ dt$$
$$=\tan x\sec^{n-2}x-(n-2)\int_0^x \sec^{n-2}t\tan^2 t\ dt$$
$$=\tan x\sec^{n-2}x-(n-2)\int_0^x \sec^{n-2}t\left(1+\sec^2 t\right)\ dt$$
$$=\tan x\sec^{n-2}x-(n-2)\int_0^x \sec^n t\ dt+(n-2)\int_0^x \sec^{n-2}t\ dt$$
$$=\tan x\sec^{n-2}x-(n-2)I_n+(n-2)I_{n-2}$$

$$I_n+(n-2)I_n=\tan x\sec^{n-2}x+(n-2)I_{n-2}$$
$$(n-1)I_n=\tan x\sec^{n-2}x+(n-2)I_{n-2}$$
$$\therefore I_n=\frac{\tan x\sec^{n-2}x}{n-1}+\left(\frac{n-2}{n-1}\right)I_{n-2}$$

ii. $\displaystyle\int_0^{\frac{\pi}{3}}\sec^4 t\ dt=\frac{\tan\,\frac{\pi}{3}\sec^2\,\frac{\pi}{3}}{3}+\frac{2}{3}\int_0^{\frac{\pi}{3}}\sec^2 t\ dt$

$$=\frac{\sqrt{3}\times 4}{3}+\frac{2}{3}[\tan t]_0^{\frac{\pi}{3}}$$
$$=\frac{4\sqrt{3}}{3}+\frac{2}{3}\left(\sqrt{3}-0\right)$$
$$=2\sqrt{3}$$

16. Calculus, EXT2 C1 2011 HSC 8a

$$I_m = \int_0^1 x^m \left(x^2-1\right)^5 dx$$

$$u = x^{m-1} \qquad u\prime = (m-1)x^{m-2}$$

$$v\prime = x\left(x^2-1\right)^5 \qquad v = \frac{1}{12}\left(x^2-1\right)^6$$

♦♦♦ *Mean mark 24%.*

$$\begin{aligned} I_m &= \left[x^{m-1} \times \frac{1}{12}\left(x^2-1\right)^6\right]_0^1 - \int_0^1 \frac{1}{12}\left(x^2-1\right)^6 \times (m-1)x^{m-2} dx \\ &= [0-0] - \frac{m-1}{12}\int_0^1 x^{m-2}\left(x^2-1\right)^6 dx \\ &= -\frac{m-1}{12}\int_0^1 x^{m-2}\left(x^2-1\right)^5\left(x^2-1\right) dx \\ &= -\frac{m-1}{12}\int_0^1 x^m\left(x^2-1\right)^5 dx + \frac{m-1}{12}\int_0^1 x^{m-2}\left(x^2-1\right)^5 dx \\ &= -\frac{m-1}{12} I_m + \frac{m-1}{12} I_{m-2} \end{aligned}$$

$$\begin{aligned} 12I_m + mI_m - I_m &= (m-1)I_{m-2} \\ (m+11)I_m &= (m-1)I_{m-2} \\ \therefore I_m &= \frac{m-1}{m+11} I_{m-2} \end{aligned}$$